Breeding Experiments on the Inheritance of Acquired Characters.[1]

By Dr. PAUL KAMMERER, University of Vienna.

ALMOST a quarter of a century has passed since I commenced to examine the inheritance of certain breeding- and colour-adaptations which I had obtained with amphibia and reptiles. I did not expect, in relatively so short a time, to obtain positive results, and, moreover, I was then well under the spell of Weismannism and Mendelism, which both agree that somatic characters are not inherited.

In the year 1909 I succeeded in ascertaining that *Salamandra atra* and *Salamandra maculosa* can be so bred as to produce a complete and hereditary interchange (of reproductive characters). The fact that *Salamandra atra*, which propagates itself in a highly differentiated manner, can be made to propagate itself in the manner of *Salamandra maculosa* need not necessarily be regarded as the acquisition of a new character, but may be an atavism. Since, however, the breeding habits of *Salamandra maculosa* can be changed to those of *Salamandra atra*, this objection is (in this case) excluded. I have hitherto always believed that no true inheritance underlay this phenomenon, but only the appearance of heredity (*Scheinvererbung*)—the external conditions applied (such as moisture) affect the germ plasm in the direct physical and not primarily physiological manner.

In view of my researches on the change of colour in *Salamandra maculosa* I could no longer entertain this belief. If the young animals are kept on a black background they lose much of their yellow marking and, after some years, appear mainly black. The offspring of these, if kept again in black surroundings, bear a row of small spots, chiefly in the middle line of the back. If the offspring, however, unlike their parents, are reared on a yellow background, these spots fuse to a band.

The yellow markings of the parent generation reared in yellow surroundings increase at the expense of the black colour of the Salamandra. If now the descendants of such strongly yellow individuals be kept on a yellow background, the yellow portions grow and appear as wide bilateral stripes. Descendants, however, which, unlike their parents, are now kept on a black background have less yellow, but proportionately far more than the background produces in the offspring of parents raised in black surroundings. The yellow markings are arranged symmetrically in rows of spots on both sides of the body.

It could now be said that the diminution of that colour which in the parents has become increased exhibited the nature of a non-inheritance. The acquired colour does not remain constant but diminishes. Ultimately, the grandchildren would have regained the same colour distribution as that of the initial parents. Therefore it could be argued we have merely an *after-effect* and *not* inheritance.

Against this view, however, we have to consider several points. (1) Young Salamandra kept on a black background, and reared from parents which had not been kept in yellow surroundings, become blacker in a much shorter time than those (on a black background) which had been reared from parents kept in yellow surroundings. (2) The descendants in my experiments are not merely placed in intermediate conditions (for example, a mixture of yellow and black backgrounds), as was done in most other breeding experiments on the inheritance of acquired characters—for example, those of Standfuss and Fischer on butterflies and of Sumner on mice. But the descendants are placed in opposite conditions: strongly yellow Salamandra are placed on a black background, and *vice versa*. Each tendency must be neutralised by the opposing stimulus; it cannot be thought that living matter behaves in this respect differently from non-living matter. (3) The yellow colour of the Salamandra, which descends from parents which have become very yellow, has at first a tendency to increase in spite of the opposing effect of black surroundings. The rows of spots of the freshly metamorphosed animals tend to fuse into stripes, just as in the case of animals brought up on a yellow background. Only later these stripes break up into spots again.

Curt Herbst, in 1919, reproached me by saying that I did not mention the augmentation of yellow pigment on a black background, and *vice versa*. It can be seen from my slides, which I have already shown in 1909 to the Congress of German Naturalists in Salzburg, and in 1910 to the International Zoological Congress in Graz, that I have made this augmentation clear. I have always emphasised this phenomenon of inheritance, which Herbst did not recognise as such.

Finally, we must not neglect how I have selected my material. For the experiments on a black background I used Salamandra which were richly marked with yellow; for those on a yellow background, Salamandra which were least marked with yellow. I used, therefore, a negative—or contra—selection to exclude the objection that I was using animals specially suitable for the colour changes which they had to undergo. I had, indeed, to contend with the fact that my animals were specially unsuitable for the experiments. Those which would have to change their colour to black were apparently burdened with a tendency to yellow, while the others which would have to produce a yellow race would have to contend against an opposing inheritance influence.

In the Vienna woods, where I had myself collected my experimental material, there were only asymmetrically spotted Salamandra (*forma typica*). My breeding experiments had changed the spotted Salamandra into the striped form. The striped race (*forma taeniata*) occurs also in the open; it is true, not in Vienna, but in districts where the earth is coloured yellow or yellowish red. In the experiments which I am about to describe I used Salamandra from the Harz Mountains, and it was found that the young of these (as in the experiment) immediately after metamorphosis already possessed their *taeniata* markings. In another case, that of Salamandra originating from the surroundings of Heidelberg, the freshly metamorphosed young were irregularly spotted, as in *forma typica*, and only arranged their markings during their growth into *taeniata*. Curt Herbst has noticed this ontogenetic recapitulation,

[1] Lecture delivered before the Cambridge Natural History Society on April 30.

and has therefore unwittingly confirmed my breeding experiments. The development of *typica* into *taeniata* is reversible, for it also happens that *forma taeniata* will change back to *forma typica*. Mr. E. G. Boulenger has confirmed this in experimental animals which he kept in the larval condition on colour backgrounds. He obtained in this way results far more beautiful and significant than my own.

At the end of the experiments, then, I have two types of striped Salamandra: first, the Salamandra which are found in Nature; and, secondly, those which have been bred in the laboratory from spotted parents. The former is an anciently established natural race, the latter a "new" laboratory race, and both of these are externally identical. I used both types for inter-crossing and inter-transplantation and also to complete my transmutation experiments.

If spotted Salamandra be crossed with naturally striped Salamandra, the offspring are of either one race character or the other in the Mendelian fashion. Spottedness is dominant over stripedness. If one crosses naturally spotted Salamandra with experimentally striped Salamandra the hybrids are of an intermediate character (stripe-spottedness) and Mendelian segregation does not occur. The hybrid indicates therefore a difference between "old" and "new" characters, even though it happens that externally both are identical. Doubtless both are heritable, but only the long-established race characteristic obeys the Mendelian laws. The new characteristic does not exhibit any atavistic tendency toward the grandparent race. These facts acquire a special interest when we recall that the vast majority of Mendelian experiments has been done on long-established race characters.

These old and new characteristics can be distinguished, not only by means of crossing-experiments, but also by means of experiments on ovarian transplantation. If ovaries of spotted females are transplanted into the naturally striped ones, then the appearance of the young is determined by the origin of the ovaries—according to the *true* mother and not according to the foster-mother. They are always irregularly spotted. If, on the other hand, ovaries of spotted females are transplanted into artificially striped ones, then, if the father is spotted, the young are line-spotted; if the father is striped, the young are wholly striped.

The ovary of the spotted female brings into the body of the naturally striped foster-mother only its own hereditary properties as effective in fertilisation. In the body of an artificially striped foster-mother this same ovary behaves as if it had been derived from the body of a striped female and as if the eggs of the striped female had been used in the crossings.

The objection cannot be raised that the operation was not thorough—that portions of the original ovaries may have been left behind in the foster-mother, as in Guthrie's experiments on fowls, which were afterwards tested by Davenport and found to be merely cases of regeneration of the original ovaries. Thanks to its enclosing membrane, the ovary of the Salamandra can be removed from the surrounding tissue as a whole. It is impossible that any remnants could have been left behind and that the descendants were derived from these remnants regenerated.

These experiments on ovarian transplantation first led me to consider the possibility of the true inheritance of somatic characters. This conception of mine was supported by the experiments of Secerov, who, to begin with, had obtained analogous changes in *Salamandra maculosa* (*forma taeniata*) by influencing the larvæ. Secondly, Secerov had measured the amount of light which was able to penetrate the interior of the body of the Salamandra. Only one-sixth of one per cent. of the outer light reached the ovaries, and the colours of the surroundings were reduced by absorption by the skin. It is improbable, therefore, that there could have been any direct colour influence on the cells of the ovary and a colour-adaptation by "parallel induction." After considering this, together with the results of transplantation, only one plausible hypothesis remains, namely, that the colour changes become inherited by a "somatic induction"; by a process similar to, if by no means the same as, that which Charles Darwin had already imagined in his theory of Pangenesis and Cunningham and Hatschek brought forward later on to explain the phenomena of heredity.

The different reactions of old and new, inherited and acquired, characters in transplantation and crossing, I have tried to make intelligible by an analogy, I must confess, provisional and crude. A new piece of clothing irritates, but this irritation diminishes the longer the clothing is worn, and it ultimately disappears. Likewise, there is a morphological irritation from each part of the body, and this diminishes in the same way. When there have been recent changes the irritation is stronger. Under suitable conditions of duration and intensity the irritation penetrates to the germ plasma. There it renders permanent a potentiality for repetition of the actual change which brought it into play. In a new character, as time goes on, the morphological irritation diminishes. Its germ-plasmic induction is no longer effective. It now belongs to the past. For the present it is no longer necessary, because without it the corresponding tendency is fixed in the germ cells. The *inductive dependence* is a relation existing between the germ plasm and only newly acquired characters. Between germ plasm and old characters, the morphological irritation of which has by use long since disappeared, there exists a complete independence as demanded by Weismann's theory and proved by Mendel's experiments.

I will now touch briefly on further results of my experiments, though these now deal not with inheritance but with changes induced on one generation. Since it is just these experiments which are cited as evidence against the inheritance of acquired characters, it will not be out of place to give a brief refutation of this, as I think, mistaken interpretation. I have succeeded in developing the rudimentary visual organ of Proteus unto a full-sized functioning eye by red illumination to which the animals were exposed for five years from birth. The degeneration of the eyes in cave-dwelling animals, according to the other view, cannot be made hereditary. It is only a non-hereditary modification, a mere environmental change. Otherwise it would be impossible, by exposure to light for a single generation, to undo what life in darkness for so many generations had produced.

What contradicts this view is that exposure to ordinary daylight is not effective. In daylight the skin which covers the rudimentary eye is filled with a dark pigment. This considerable but by no means complete absorption of light due to the pigment is sufficient to arrest the development of the eye, so that the normal degeneration occurs. Red light, however, causes no pigmentation in the skin, and only under the influence of this chemically inactive light is the regression overcome.

The misinterpretation of these data allows me to make a further general observation. In order to prove completely that acquired characters are inherited we must produce at least one alteration of an inborn property. But if we only recognise those properties as hereditary which are unchangeable, then we have *from the very outset* excluded all heritable transformations, and at the same time rendered useless any investigations of the matter. If that which changes cannot be hereditary, and if that which is hereditary cannot change, we can only predict the immutability of species and therewith dogmatically leave on one side, not only the inheritance of acquired characters, but also the whole theory of evolution.

All existing objections, which rendered insoluble the inheritance of acquired characters, apply also to my breeding experiments on Alytes, and I myself would not have attached any special significance to this were it not that it is a result of just these experiments which has aroused the keenest interest in England—the development of a nuptial pad in the male Alytes. In male frogs, which pass their mating-time in water, there appears before mating, usually on the inner fingers, a rough, horny, glandular, dark-coloured pad. On the other hand, in Alytes, which mate on land, no trace of such a pad is to be seen. Yet it can be made to appear after several generations by compelling the Alytes to mate in water, like other European frogs and toads. This compulsion is brought about by raising the temperature, under which condition the mating animals stay longer in the water than usual, for if they did not do so they would run the risk of being dried up. Later in life compulsion becomes unnecessary. The stimulus of warmth produces an association through which henceforward the Alytes take to the water of their own accord when they wish to mate.

Of the many changes which gradually appear in this water breed during the various stages of development—egg, larva, and the metamorphosed animal, young and old—I will describe only one, the above-mentioned nuptial pad of the male. At first it is confined to the innermost fingers, but in subsequent breeding seasons it extends to the other fingers, to the balls of the thumb, even to the underside of the lower arm. After spreading, it exhibits an unexpected variability, both in the same individual and between one individual and another. The variability in the same individual is shown by the characters altering from year to year and in the absence of symmetry between the right hand and the left. In one specimen the dark pad extended to all the other fingers and almost over the whole of the left hand. On the right hand it was never so marked, and it was even less developed later, because the skin was stripped from this hand in the living animal for the purpose of histological investigation. The present skin and pad formation next to the inner finger is to be ascribed to regeneration in the mating season which followed. Microscopical preparations show the difference between the thumb skins of the mating male Alytes in the control breed and the padded skins of the water breed. The skin on the thumb of the normal male is subject likewise to an annual change in thickness. Alytes has already in its natural state a tendency to pad formation, and therefore does not display such a striking novelty as microscopic observation would lead one to think.

The great variability and extent of the pad, which can be produced by cultivation, and its independence of the testes, as castration experiments show, render the hypothesis possible, that what we are dealing with is an artificial creation of a new function. On the other hand, the Alytes pad can be interpreted as an atavism; or again, since the tendency was already there, one can quite well deny that the character has been acquired. Further, the influence of the heat responsible for the change penetrates the whole body of the cold-blooded animal and may therefore penetrate to the germ plasm in a purely physical manner. It is true that when four generations have altered in a similar manner, even after the stimulus has been removed, it is not very plausible that parallel induction should be the cause, and the subsequent appearances a mere after-effect. But as the atavism objection can always be raised, it is not very clear to me why just this experiment (with Alytes) is so often looked upon as an *experimentum crucis*. In my opinion it is by no means a conclusive proof of the inheritance of acquired characters.

Not content with any of the previous experiments, I carried out, before 1914, what may really be an *experimentum crucis*. I have written a few words on it in my "Allgemeine Biologie." There has been no detailed publication as yet. The subject is the Ascidian, *Ciona intestinalis*. If one cuts off the two siphons (inhalent and exhalent tubes), they grow again and become somewhat larger than they were previously. Repeated amputations on each individual specimen give finally very long tubes in which the successive new growths produce a jointed appearance of the siphons. The offspring of these individuals have also siphons longer than usual, but the jointed appearance has now been smoothed out. When nodes are to be observed, they are due not to the operation but to interruptions in the period of growth, just as in the winter formation of rings in trees. That is to say, the particular character of the regeneration is not transferred to the progeny, but a locally increased intensity of growth is transferred. In unretouched photographs of two young Ciona attached by their stolons to the scratched glass of an aquarium, the upper specimen is clearly seen to be contracted; the lower is at rest and shows its monstrously long siphons in full extension. They were already there at birth, for it was bred from parents the siphons of which had become elongated by repeated amputation and growth.

In those animals with artificially lengthened siphons we can, furthermore, combine with the amputation at the front end another amputation at the hinder end. At the hinder end—in the coils of the intestine

—lies the generative organ, an hermaphrodite gland. We remove the whole of this part of the body and leave the front part to regenerate and to reproduce a new generative organ: that is, *new germ plasm is formed from somatic tissue.* It has been established already in several species of animals and plants that Weismann's "continuity of germ plasm" is not obligatory but, at most, a facultative continuity. The long-siphon Ascidians with regenerated germ plasm give birth to progeny also long-siphoned. In this way the most familiar objection brought against the inheritance of acquired characters—the claim that there is a direct influence on the germ plasm—is, I think, definitely removed. The local character of the operation in cutting off the siphons renders this chief objection almost inapplicable. We might, however, still argue that physical influence still obtains; that while I am cutting the siphons at the head, a direct physical reaction is taking place on the germ plasm. In this case there would already be established that tendency which would give rise to an apparent inheritance in the progeny.

But now we cut away all the generative organ, with all its germ cells and its active and latent tendencies. We await the growth of a new generative organ. The regeneration takes place at a time when there are no further disturbances influencing the body. Nevertheless, the growth to which it gives rise is still affected. The change therefore cannot have been lying preformed in the original germ plasm. It can have come ultimately from nowhere but from the changed body.

The present circumstances are scarcely favourable for the furtherance of these researches in heredity in my impoverished country. During the War experimental animals, the pedigrees of which were known and had been followed for the previous fifteen years, were lost. I am no longer young enough to repeat for another fifteen years or more the experiments, with the results of which I have been long familiar, before I attempt to break new ground. The necessities of life have almost compelled me to abandon all hope of pursuing ever again my proper work—the work of experimental research. I hope and wish with all my heart that this hospitable land may offer opportunity to many workers to test what has already been achieved and to bring to a satisfactory conclusion what has been begun.

The Earth's Electric and Magnetic Fields.[1]

By Prof. W. F. G. SWANN, University of Minnesota.

I.

QUITE apart from those more spectacular manifestations of atmospheric electric phenomena associated with the thunderstorm, we have to recognise the following facts, as pertaining to the ordinary quiet day:

(1) The earth is charged negatively to such an extent as to give rise to a vertical potential gradient which amounts to about 150 volts per metre at the surface of the earth, and goes through fairly regular variations throughout the day and throughout the year, variations amounting to 50 per cent. or more of its total value.

(2) The potential gradient diminishes with altitude until its value at 10 kilometres is practically negligible compared with that at the earth's surface, a result which is brought about by the existence, in the atmosphere, of a positive charge, the total amount of which below the altitude 10 kilometres is practically equal to the negative charge on the earth's surface.

(3) The atmosphere is a conductor of electricity. The conductivity near the earth's surface is so small that a column of the air one inch long offers as much resistance to the flow of the electric current as would a copper cable of equal cross section extending to the star Arcturus and back twenty times over.

(4) In spite of the smallness of the conductivity of the atmosphere at the earth's surface, its amount is nevertheless sufficient to ensure that 90 per cent. of the earth's charge would disappear in ten minutes if there were no means of replenishing the loss.

(5) The conductivity increases with altitude at such a rate that its value at an altitude of 10 kilometres is about fifty times that at the earth's surface, and there is indirect evidence to substantiate the belief that at altitudes of the order of 100 kilometres it may attain a value more than 10^{11} times that at the earth's surface. Such a conductivity would cause the upper atmosphere to act, practically, as a perfect conductor in its relation to phenomena in the lower atmosphere.

(6) There is some evidence for and some against the view that our atmosphere is traversed by a radiation of cosmical origin, and of penetrating power ten times, or more, that of the gamma rays of radium.

A potent factor contributing to the conductivity of the atmosphere is the radioactive material in the air and soil. There are, on the average, about 1.5 molecules of radium emanation per c.c. of the atmosphere over land, yet this small amount is sufficient to contribute very appreciably to the ionisation there. On the basis of the known amounts of radium and thorium emanations in the atmosphere, and of radioactive materials in the soil, we could account fairly well for the ionisation of the lower atmosphere. The conductivity of the air over the great oceans is, however, practically as great as it is over land, and is very much greater than can be accounted for by the radioactive materials, which are negligible in amount in the ocean and in the air over it. The assumption of a penetrating radiation would provide a cause for the ionisation known to exist over the sea. If, however, we are unwilling to admit the existence of such a radiation, the ionisation over the ocean remains to some extent a mystery, and may have to be attributed to a small spontaneous ionisation of the gas.

The great problem of atmospheric electricity is, of course, the explanation of the maintenance of the earth's charge. The replenishment to be accounted for is small, amounting to only 1000 amperes for the

[1] Portions of a lecture on "Unsolved Problems of Cosmical Physics," delivered before the Franklin Institute on December 30, and published in full in the Journal of the Franklin Institute.